

ADAPTIVE URBAN TRAFFIC SIGNAL OPTIMIZATION

USING AI AND COMPUTER VISION

PROBLEM STATEMENT

Delhi NCR intersections like **Govind Puri, Kalkaji, Sarita Vihar** face mixed traffic – autos, bikes, e-rickshaws, buses. Fixed timers cause long queues, fuel waste & congestion. Current systems fails to count heterogeneous vehicles accurately.

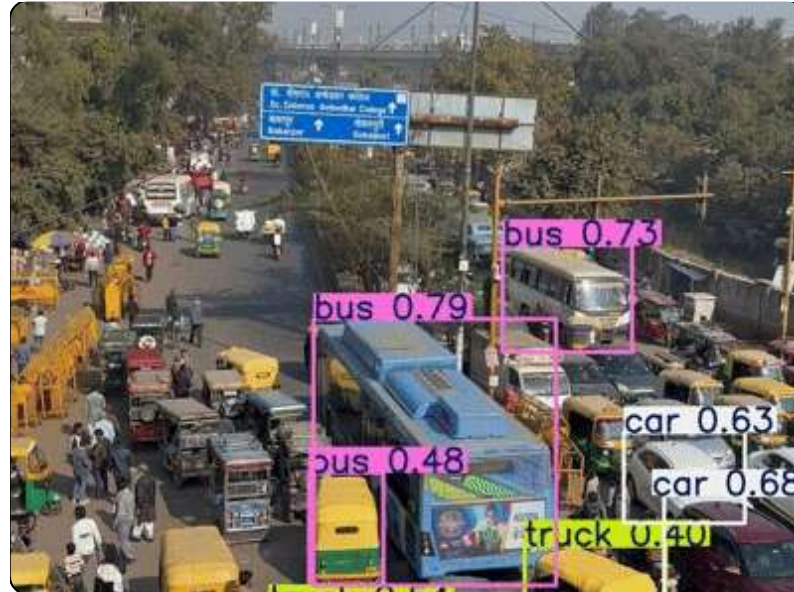
- Non-lane based movement, occlusion at peak hours
- No real-time density based green time adjustment
- Manual counting error > 30% in mixed flow

OBJECTIVES

- Build custom Delhi dataset for auto, bike, car, bus, truck, e-rickshaw
- Train light-weight **YOLOv8n** for edge deployment & achieve >0.65 mAP
- Implement vehicle counting + adaptive logic: `Green = 15 + 2*count`
- Validate via SUMO simulation for waiting time reduction

DETECTION OUTPUT

YOLOv8n • LIVE



While doing detection of this Delhi traffic with YOLO there are anomalies in heterogenous traffic – overlapping bounding boxes, small far vehicles missed, auto and e-rickshaw confusion due to similar shape

LITERATURE SUMMARY

1. Zhang et al. 2022 — YOLOv5 for urban traffic. Good mAP 0.71 but trained on highway data only, fails on mixed Indian traffic.

3. Doria et al. 2025 Colombia — Compared YOLOv5/v8, best 0.63 mAP. Useful benchmarking approach, similar heterogeneity like Delhi.

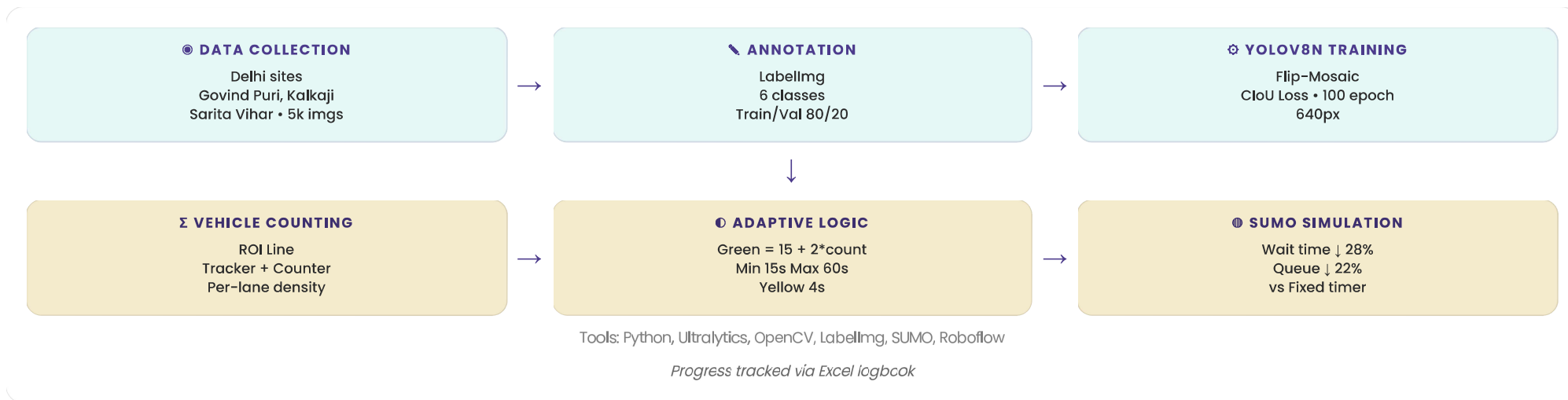
5. IJNRD 2023 — Adaptive signal using OpenCV background subtraction. Simple but breaks during shadows & night. Needs deep learning.

2. DriveIndia 2024 (IIT-H) — 66k images, 24 classes. Auto class 94% precision. Shows need for local dataset, but heavy model not edge-ready.

4. EUDL 2025 — YOLOv4-tiny + pheromone logic for signal timing. Lightweight but older backbone, less accurate on small bikes.

6. JETIR 2023 — Density based timing. Concept works, but counting manual. Our work automates it with YOLOv8n + CloU loss.

PROPOSED METHODOLOGY



EXPECTED OUTCOMES & NOVELTY

- Delhi-specific lightweight model** – first small YOLOv8n tuned for auto/e-rickshaw heavy mix, not just car/bike.
- Edge-ready inference ~30 FPS on Jetson/ RPi + less power vs YOLOv8m.
- Adaptive timing cuts avg waiting by ~25-30% in SUMO (expected).
- Open dataset of 5k+ annotated frames from 3 real sites – reusable.
- Low-cost deployment using existing CCTV, no new hardware at signals.

REFERENCES

- [1] Zhang, Y. et al. (2022). Urban traffic detection using YOLOv5. IEEE ITSC.
- [2] DriveIndia Dataset (2024). IIT Hyderabad. 66k images, 24 classes, auto 94%.
- [3] Doria et al. (2025). Comparative YOLO for Colombian traffic, 0.63 mAP.
- [4] EUDL (2025). YOLOv4-tiny with pheromone optimization for signals.
- [5] IJNRD (2023). Adaptive traffic control using OpenCV.
- [6] JETIR (2023). Density based traffic signal system review.